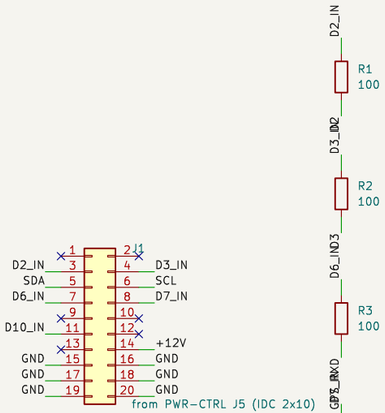
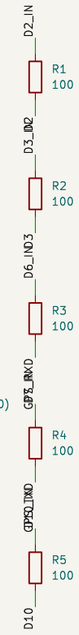
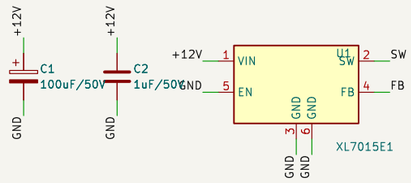
1. FROM PWR-CTRL J5 (2x10 ribbon, pin 1 = red stripe)



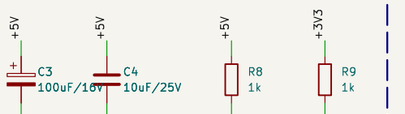


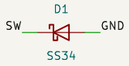
Not connected on purpose: 1 ACC_SENSE, 2 HOLD (pulling it low cuts the PC power), 9/10 vehicle bus, 12 main 3V3, 13 5V_EXT.
The hub makes its own 5 V / 3.3 V from pin 14 (VSW, 12 V, fused 1.1 A on the main board).
100R series resistors protect the XIAO pins against wiring mistakes.

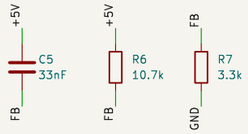
2. 5.3 V BUCK (XL7015) + 3.3 V LDO

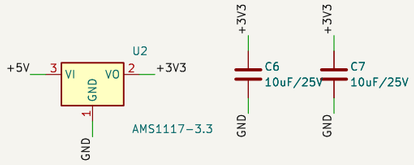










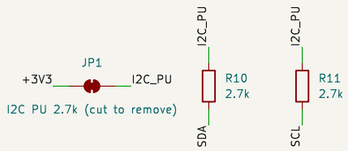




5 V: 5.30 V, up to ~0.7 A total (XL7015). 3.3 V: AMS1117, up to ~0.4 A (1 W dissipation limit).
These rails are separate from the XIAO 3V3 / main 5V_EXT, so sensors do not load the MCU supply.



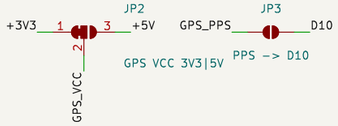
3. I2C PORTS (400 kHz). 4x Qwiic 3.3 V + 2x XH4 with 5 V power

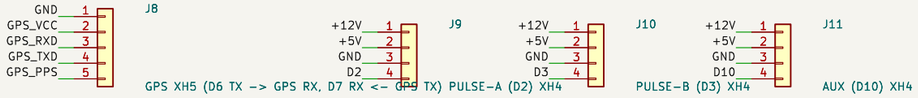




Pull-ups: 2.7k here || 4.7k on main (JP3) = 1.7k, fine for 400 kHz.
XH4 I2C ports: 5 V power, 3.3 V logic (use modules with their own regulator / level shifter).

4. GPS (UART) / PULSE / AUX





PULSE-A/B and AUX have the same pin order as PWR-CTRL J8/J9 (1 12V, 2 5V, 3 GND, 4 IO); PULSE-IN plugs in with a straight cable.
D2/D3 are also on main J8/J9: use only one connector per pin. D10 is shared by AUX and GPS PPS (JP3): use one of them.
JP2 selects GPS power (default 3V3).

OpenCruise

Sheet: /
File: exthub.kicad_sch

Title: OpenCruise EXT-HUB (port hub for PWR-CTRL J5)

Size: A3	Date: 2026-11-09	**Rev: 1.0**
KiCad E.D.A. 9.0.9		Id: 1/1